

# VISTRA

Vision-based, Self-aware, Traversability- and Risk-aware Autonomy

## PS 26126

Vision-based autonomous navigation for an outdoor UGV

## CORE THESIS

Uncertainty from perception -> risk -> planner  
-> behavior

## EVIDENCE

Seeded simulation + baseline + fault injection  
+ replay

## SOURCE INTEGRITY

PS-specific content: supplied VISTRA blueprint. External facts: trusted source register. No unmeasured performance claims.

# 1 | Problem & solution

Target: explainable, uncertainty-aware outdoor navigation

## PS REQUIREMENTS

Lightweight perception  
Visual odometry / SLAM  
Planning + wheel/motor commands  
Collision-free A -> B

## CONVENTIONAL GAP

Detector-only perception can miss untrained hazards  
Hard binary maps hide model uncertainty

## VISTRA RESPONSE

Free-space-first risk map  
NCS behavior gate  
TTC safety override

## EVIDENCE LOOP

Baseline vs VISTRA on identical seeds; fault injection for blur/dark/obstacle; structured logs; measured FPS/latency; scenario-level statistics

## IMPORTANT

This deck contains no fabricated VISTRA performance numbers.

## 2 | Architecture

Camera-first closed loop

### 1. PERCEPTION

Camera  
SegFormer-B0  
Dynamic-object tracking  
Input quality

### 2. MAP

IPM  
Traversability  
Entropy  
BEV risk + fusion

### 3. LOCALIZE

Stereo VO/SLAM  
EKF  
Pose + covariance

### 4. PLAN

D\* Lite  
DWA  
Goal + risk + clearance

### 5. SAFETY

TTC  
NCS  
Recovery  
Wheels

### FEEDBACK LOOPS

Pose covariance inflates risk-map costs; NCS caps speed; localization failure invokes Recovery Manager; planning failure escalates to replan / rotate-search / safe stop

### DASHBOARD

Camera overlay | BEV map | pose/path | NCS | TTC | state/e-stop | structured explanation | replan markers | wheel commands | fault injection | baseline comparison

# 3 | Risk-aware traversability

Probabilities are retained instead of collapsed to one class

## CORE FORMULAE

$$T = \sum(p_k * t_k)$$

$$U = \text{normalized entropy}(p)$$

$$R = (1-T) + \lambda_u U + \lambda_g G + \lambda_{\text{prox}} P$$

## BEV

Example: 20 m x 20 m at 0.2 m

= 100 x 100 cells

Temporal log-odds / decay

## 6 CLASSES

Smooth-drivable; Rough-drivable; Vegetation/non-traversable; Hard obstacle; Water/ditch/negative; Background/sky

## RISK BANDS

Green <0.25

Yellow <0.5

Orange <0.75

Red >=0.75 or lethal

## LIMITATION

IPM assumes locally flat ground. Proposed mitigation: IMU pitch/roll compensation and lower weight for far-range cells.

[SegFormer paper](#)

[SegFormer docs](#)

# 4 | Localization & recovery

Stereo vision remains primary; degraded states are explicit

## PRIMARY PATH

Stereo VO/SLAM + loop-closure  
RTAB-Map or ORB-SLAM3  
IMU/wheel EKF bridges short gaps

## CONFIDENCE SIGNALS

Features; RANSAC inliers; reprojection error; EKF covariance; blur;  
brightness; feature distribution

## NORMAL

>0.7  
nominal speed cap

## CAUTION

0.4-0.7  
slow + inflate

## RECOVER

0.2-0.4  
relocalize/re-observe

## SAFE\_STOP

<0.2  
hold + alert

## RECOVERY SEQUENCE

confidence fall -> slow -> stop -> rotate-search -> retrace to last good keyframe -> resume after confidence restoration

[ORB-SLAM3 paper](#)

[ROS 2 Jazzy](#)

# 5 | Planning + predictive safety

Global route, local motion and a reactive override

## GLOBAL

D\* Lite  
Incremental replanning on risk grid

## LOCAL

DWA  
Reachable (v,w)  
progress + risk + clearance

## TTC

3 s horizon  
0.1 s steps  
constant-velocity obstacle model

## STOPPING TIME

$\tau_{\text{stop}} = t_{\text{latency}} + v/a_{\text{max}}$   
Latency is measured end-to-end.  
Planner mapping:  
 $v_{\text{left}} = v - wL/2$   
 $v_{\text{right}} = v + wL/2$

## STARTING TTC RULES

$>3*\tau_{\text{stop}}$ : continue  
 $1.5*\tau_{\text{stop}}-3*\tau_{\text{stop}}$ : slow + replan  
 $\tau_{\text{stop}}-1.5*\tau_{\text{stop}}$ : hard brake + replan  
 $\leq \tau_{\text{stop}}$ : emergency stop

[D\\* Lite](#)

[DWA](#)

[Gazebo Harmonic](#)

# 6 | Data & datasets

Public data for research; held-out real data for validation

## RUGD

7,453 pixel-wise labeled images  
Unstructured outdoor robot driving  
Irregular boundaries + blur challenges

## RELLIS-3D

13,556 LiDAR scans  
6,235 images  
20-class ontology  
Multimodal off-road data

## SEGFORMER-B0

MiT-B0: 3.7M parameters in HF docs  
Project FPS must be measured

## TRAIN / VALIDATE

Sequence/location-level splits  
Held-out real set never used in training  
Hard negatives: shadows, puddles, grass hiding rocks, glare

## PROJECT DATA

Blueprint plans about 300-500 campus/trail frames  
This is a plan, not a completed result.

## INTEGRITY

Never import paper metrics into VISTRA result tables.

[RUGD](#)

[RELLIS-3D](#)

[ONNX Runtime](#)

# 7 | Benchmark protocol

Evidence comes from repeated seeded experiments

## SCENARIOS

11 families: terrain, obstacles, lighting, localization, perception, narrow paths, dead ends, route choice, unknown obstacles

## RUNS

20-30 seeded runs per scenario per system  
Mean + confidence interval

## BASELINE

YOLOv8 + same VO/SLAM + A\* + fixed-distance stop

## METRICS

Success; collisions; path length; time; localization error; path deviation; replans; FPS; latency; NCS trace; recovery time

## ABLATIONS

A: remove uncertainty term  
B: remove confidence governor  
C: remove TTC/terrain safety

## RULE

If the baseline wins a scenario, report it. Do not fabricate or hide results.

## 8 | Dashboard & explainability

The system should show not just what it does, but why

### PERCEPTION VIEW

Camera feed + segmentation overlay

### MAP VIEW

BEV risk map + global/local path + actual path  
+ pose

### SAFETY VIEW

NCS + components + TTC + speed + state/e-stop

### EXPLANATION FEED

time | state | trigger | confidence | speed cap | chosen action

### INTERACTION

Spawn obstacle; blur; darken; camera loss; baseline comparison; replan markers

### DATA FLOW

FastAPI/WebSocket -> React/TypeScript -> 2D canvas/SVG. Offline starter included in Source-Code/Dashboard.

[FastAPI](#)

[CVAT](#)

# 9 | Risks, limits and roadmap

Build the core first; protect the benchmark

## HIGH RISKS

- SLAM integration time
- Sim-to-real gap
- Ditch recall
- Latency too high for TTC
- Scope creep

## MITIGATIONS

- Time-box SLAM
- Separate real-video track
- Uncertainty + geometry
- Measure end-to-end latency
- Freeze scope

## MUST-HAVE

- Simulation + vehicle interface
- Segmentation + BEV risk
- VO + EKF
- D\* Lite + DWA
- TTC + dashboard + benchmark

## IMPORTANT

- NCS
- Recovery Manager
- Decision log
- Fault injector

## ADVANCED / OPTIONAL

- Relative depth
- Moving-object prediction
- Live side-by-side
- Physical UGV after core

## ROADMAP

Weeks 1-6: build and freeze core. Weeks 7-10: test, benchmark and rehearse.

[Gazebo Harmonic official docs](#)

# 10 | Judge-ready demo

A simple, observable sequence

01

No GPS  
Empty map

02

Segmentation  
Risk map builds

03

Drive A -> B  
Path visible

04

Obstacle fault  
TTC + replan

05

Blur/dark  
NCS drops

06

Recover  
Relocalize

07

Reach goal  
Metrics summary

08

Baseline  
Same seed

## BACKUP

Rosbag replay; pre-rendered full run; recorded-video perception; playback dashboard

## SOURCE

# 11 | Trusted sources

Click a source label to open the official page or paper

S1 | BEL official

[OPEN](#)

S3 | Gazebo Harmonic

[OPEN](#)

S5 | SegFormer docs

[OPEN](#)

S7 | D\* Lite

[OPEN](#)

S9 | RUGD paper

[OPEN](#)

S11 | RELIS-3D paper

[OPEN](#)

S13 | CVAT

[OPEN](#)

S15 | FastAPI

[OPEN](#)

S2 | ROS 2 Jazzy

[OPEN](#)

S4 | SegFormer paper

[OPEN](#)

S6 | ORB-SLAM3 paper

[OPEN](#)

S8 | DWA

[OPEN](#)

S10 | RELIS-3D repo

[OPEN](#)

S12 | ONNX Runtime

[OPEN](#)

S14 | OpenCV

[OPEN](#)

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